



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Programming Languages



Learning Outcome:

Talk about principles, concepts, and characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- Do you think it is easy to understand binary code just by looking at it?
- What kind of training do you need in order to use high-level programming languages?

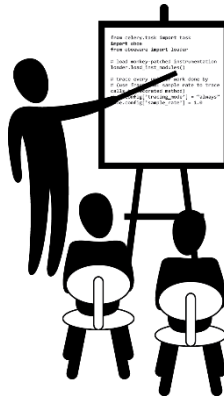
Can Do Statements:

- I can discuss characteristics of low level languages.
- I can talk about characteristics of high level languages.
- I can compare "compiled code" to "interpreted code".



Let's Explore!

Peter was late for class



Instructions: Read the text below.

- Peter:** Now I see. What about high-level languages?
- Danny:** When people use the term programming languages, they actually refer to high-level languages. Programmers might want to write programs using high-level languages, such as Java, Fortran, or C++, because they are closer to the logic of human language; you use words and symbols, and you ought to use a specific syntax depending on the language. Additionally, you might want to use a high-level language code to run it on different hardware because it is portable.
- Danny:** Hi, Peter! Where are you going?
- Peter:** Hi! I'm meeting Mr. McGill in an hour. He's going to explain to me what high-level and low-level languages are. Tomorrow, I'm giving an oral presentation about it, and I'm a bit confused.
- Danny:** Don't worry. Actually, I'm taking an exam at the faculty later! I'll walk with you and tell you what I know about it.
- Peter:** That'd be great!
- Danny:** Well, as you know, there are two types of programming languages: high-level and low-level languages. Low-level programming languages are machine-oriented as they're used to communicate directly to computer hardware. There are two types of low-level languages: machine language and assembly language. Machine language is absolutely difficult for humans to understand as it is written in binary code; however, assembly language, which uses English-like instructions, makes machine code more readable by programmers. You ought to use a translator program called assembler to translate the assembly language into machine language.
- Peter:** What if I need to program a piece of hardware? Can I use a high-level language?
- Danny:** Actually, you'd better use a compiler or an interpreter. These computer programs translate high-level language to machine code, just that they do it differently.
- Peter:** How's that?
- Danny:** Well, a compiler translates the code before you use the program. You ought to compile the code first before running it. This way, the code will run faster. You might want to use a compiler if you need to run the program repeatedly as the compiler will give you a file with an .exe extension. On the other hand, an interpreter will translate the code line during the execution; it is not going to be as fast as a compiled code, but surely it'll be more flexible and interactive at the moment of translation.
- Peter:** That was such a good explanation. I'd better call Mr. McGill and tell him I won't need his help. Thanks a lot!

**Let's Practice!**

Low level language basics

Instructions: Read the sentences. Write the missing word in the blank. Use: uses, has, manages, symbols and using.

1. The programmer is _____ low level language to communicate directly to the machine.
2. The two categories that low level language _____, are machine oriented and assembly.
3. Machine oriented category _____ binary code.
4. Assembly language communicates with _____ and letters.
5. Low-level language _____ computer's hardware components and constraints.

**Let's Practice!**

Remember the difference between compiler and interpreter

Instructions: Select the correct answer.

1. What are compiler and interpreter?
 - a. Programs to translate idioms
 - b. Programs to write scripts
 - c. Programs that translate high level language into machine code
 - d. Programs to translate Spanish to English
2. Which program translates the code at once?
 - a. Binary
 - b. Compiler
 - c. Interpreter
 - d. Seinfeld
3. Which program translates the information in real time?
 - a. Binary
 - b. Compiler
 - c. Interpreter
 - d. Seinfeld
4. Does compiler translate the code before you use the program?
 - a. No it can't.
 - b. Yes it does.
 - c. No you aren't.
 - d. Yes you should.
5. Which program is more interactive and flexible?
 - a. Binary
 - b. Compiler
 - c. Interpreter
 - d. Seinfeld





Extension Activity

There are many programming languages

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about one low level language and one high level language
- Describe and write the difference between the two languages
- Share you findings with your teacher and classmates

Self-Assessment

Yes

No

- I can investigate about low level and high level languages
- I can write paragraphs about programming languages
- I can talk about programming languages and their differences

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Target Words

- **interpreter** (noun): a computer program that translates high-level code into machine code at the moment of running it
e.g. An interpreter will translate the code line during the execution.
- **English-like** (adjective): similar to the English language
e.g. Assembly language, which uses English-like instructions, makes machine code more readable by programmers.
- **run** (verb): to execute a program or code
e.g. You ought to compile the code first before running it.
- **portable** (adjective): able to be carried or stored in a flash drive easily
e.g. You might want to use a high-level language code to run it on different hardware because it is portable.
- **Translate** (verb): to convert data, information, or codes into a different coding system
e.g. These computer programs translate high-level language to machine code.



Learning Points

The Future With the Present Continuous

Typically, you use the **present continuous** to talk about actions that are happening at the moment of speaking. Another common use of the **present continuous** is to talk about **future arrangements**.

You can use the **present continuous** to talk about **arrangements** in the **future** that you have already planned. Example: **meeting** the team of developers next week. It is recommended to mention a **time expression** that refers to the **future** to make it clear that you are talking about a **future event** instead of an action that is happening now. Example: She **is taking** a training course on high-level languages **next month**.

Examples

- *He is talking about the types of programming languages.*
- *What is Mr. Douglas explaining today?*

Compilers and Interpreters

When you talk about high-level languages, you need to know what **compilers** and **interpreters** are.

Compilers and **interpreters** are the two ways programmers use to translate high-level languages into machine codes; it means the ways computer can understand the program.

Compilers and **interpreters** are different systems that provide different advantages.

Compilers

A compiler is a program that translates the source code or high-level language into machine code all at once. This means that the translation is only done once, and it has to be done before we can execute the program in order to obtain an output.

Interpreter

An **interpreter** also translates high-level language into machine code, but it does it differently. It translates the code at the same time the program is executed, so every time you execute the program, the code translation is happening simultaneously.

Each system has its advantages. Let's see:

Compiler	Interpreter
✓ It is faster.	✓ It is more flexible and interactive.
✓ It doesn't reveal the source code.	✓ It is translated as needed.

